Portfolio Research Project

Volume Signal Analysis

Testing Trading Volume as a Predictive Factor

Research Objective

The objective of this experiment was to determine whether trading volume contains useful information for predicting future stock returns and whether volume can be used as a standalone factor in a systematic portfolio strategy.

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1. Introduction

Trading volume is one of the most commonly studied variables in financial market data. It represents the number of shares traded during a particular period and can potentially provide information about investor activity, liquidity, market attention, and the strength of price movements.

The purpose of this experiment was to investigate whether the volume information contained in the available OHLCV dataset could be transformed into a useful predictive signal.

The central research question was:

> *Does trading volume provide information about future stock returns that can be exploited systematically?*

Rather than assuming that high or low volume is inherently predictive, the analysis tested the relationship empirically by sorting observations into volume groups and comparing their subsequent returns.

The analysis ultimately showed that volume, when considered by itself, did not provide a sufficiently clear or stable predictive relationship with future returns.

However, this does not imply that volume is useless. The results suggest that volume may be more useful as a supporting variable, filter, or feature combined with other signals.

2. Dataset

The analysis was performed using historical OHLCV data.

The dataset contained the following variables:

- Date
- Stock identifier
- Open price
- High price
- Low price
- Close price
- Volume

The dataset contained:

- 35,616 total rows

- 24 stocks
- Date range from September 2016 to September 2022
- 34,656 observations available for the forward-return analysis

The presence of multiple stocks and several years of historical data allowed us to test whether the volume relationship was consistent across securities and through time.

3. Research Methodology

The volume analysis followed a simple factor-testing framework.

3.1 Step 1: Obtain Trading Volume

For every stock and trading day, the recorded trading volume was used as the volume variable.

The basic signal can be represented as

$$V_{i,t}$$

where $V_{i,t}$ is the trading volume of stock i on day t.

The initial experiment deliberately used raw volume in order to determine whether the simplest possible volume-based signal could provide useful predictive information.

3.2 Step 2: Rank Observations by Volume

Observations were ranked according to their trading volume and divided into approximately equal-sized groups.

The general structure was:

$$\text{Group } 1 \rightarrow \text{Lowest Volume}$$

and

$$\text{Group } 10 \rightarrow \text{Highest Volume}.$$

The purpose of this grouping was to determine whether future returns changed systematically as volume increased.

If volume were a useful standalone factor, we would expect some consistent relationship between the volume ranking and future returns.

3.3 Step 3: Calculate 20-Day Forward Returns

For each observation, the future return was calculated over a 20-trading-day horizon.

The forward return can be represented as

$$R_{i,t}^{(20)} = \frac{P_{i,t+20}}{P_{i,t}} - 1$$

where

- $P_{i,t}$ is the price of stock i at time t,
- $P_{i,t+20}$ is the price approximately 20 trading days later.

The purpose of this calculation was to test whether today's volume contains information about future stock performance.

3.4 Step 4: Calculate Group-Level Returns

For every volume group, the mean 20-day forward return was calculated.

The relationship being tested was therefore

$$\text{Average Forward Return} = f(\text{Volume Group}).$$

If volume were a strong standalone predictive factor, the group returns would ideally demonstrate a relatively clear pattern.

For example, an increasing relationship would appear as

$$R_1 < R_2 < R_3 < \cdots < R_{10}.$$

A decreasing relationship would appear as

$$R_1 > R_2 > R_3 > \cdots > R_{10}.$$

A highly irregular relationship would suggest that raw volume does not provide a simple monotonic predictive signal.

4. Experimental Results

The average 20-day forward returns obtained from the recorded volume-group analysis were:

Table 1: Average 20-Day Forward Return by Volume Group

Volume Group	Average 20-Day Forward Return
Group 1	0.016540
Group 2	0.015959
Group 3	0.017303
Group 4	0.016691
Group 5	0.017677
Group 6	0.020941
Group 7	0.017315
Group 8	0.015197

The available results show that the average forward returns vary across the volume groups, but they do not form a clear monotonic pattern.

For example, Group 6 has the highest observed average return among the available groups, while Group 8 has a considerably lower average return.

This is important because increasing volume does not consistently lead to increasing future returns.

4.1 Volume Group vs. Forward Return Graph

Figure 1 visualizes the relationship between volume groups and their corresponding average 20-day forward returns.

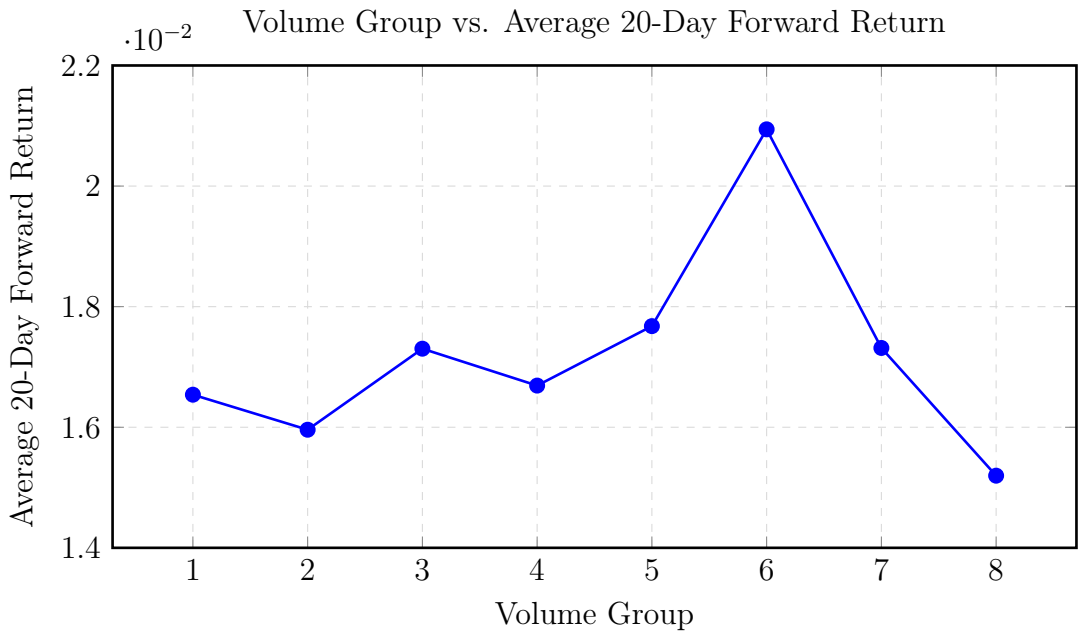


Figure 1: Average 20-day forward return across the available volume groups.

The graph provides a visual representation of the main result of the experiment.

The points fluctuate rather than following a consistent upward or downward trend. In particular, the return rises from Group 4 to Group 6 and then falls again in Groups 7 and 8.

Therefore, the relationship between raw volume and future returns appears to be non-monotonic.

5. Interpretation of Results

The experiment does not show a sufficiently strong standalone volume effect.

The main observation is that the relationship between volume and future returns is irregular.

In other words, increasing volume does not consistently correspond to increasing future returns.

Similarly, decreasing volume does not consistently correspond to a specific future return outcome.

The result can therefore be summarized as

Raw volume does not show a clear monotonic predictive relationship with 20-day forward returns.

This does not prove that volume has no relationship with returns. Instead, it indicates that the particular raw-volume ranking approach tested here was not sufficiently strong or consistent to justify treating raw volume as a standalone trading signal.

6. Why the Volume Test Failed

Several factors can explain why raw volume did not produce a strong standalone signal.

6.1 1. Raw Volume Is Not Directly Comparable Across Stocks

Different stocks naturally have very different trading volumes.

For example, a large and actively traded company may have millions of shares traded per day, while a smaller company may have a much lower absolute volume.

Therefore,

$$V_{\text{Stock A}} > V_{\text{Stock B}}$$

does not necessarily mean that Stock A is experiencing unusually high trading activity.

Raw volume contains a strong stock-specific liquidity component.

Consequently, ranking stocks directly using absolute volume may partially rank them according to their normal trading size rather than unusual activity.

6.2 2. Volume Is Highly Stock-Dependent

Volume characteristics differ substantially between securities.

Some stocks consistently have high trading activity, while others consistently trade lower quantities.

This means that a stock may repeatedly appear in a high-volume group simply because it normally trades a large number of shares.

Therefore, raw volume may not capture the information that is actually relevant for prediction.

6.3 3. No Clear Monotonic Relationship

One of the most important findings is the absence of a clear monotonic relationship.

The group returns do not consistently increase or decrease as the volume ranking increases.

For example, the observed values include:

$$0.015959, \quad 0.017303, \quad 0.016691, \quad 0.017677, \quad 0.020941, \quad 0.017315, \quad 0.015197.$$

The highest observed value occurs in the middle of the ranking rather than at the highest-volume end.

This weakens the case for using raw volume as a simple ranking factor.

6.4 4. Volume May Be a Conditional Signal

Volume may not independently predict returns but could provide useful information when combined with another variable.

For example, unusually high volume accompanying strong price momentum may have a different interpretation from unusually high volume without any meaningful price movement.

Therefore,

$$\text{Volume}$$

may be less useful on its own than

$$\text{Price Signal} + \text{Volume Confirmation}.$$

This distinction is important for the next stage of the research.

6.5 5. Raw Volume Does Not Measure Abnormal Activity

A more informative quantity may be relative or abnormal volume.

For example, relative volume can be defined as

$$RV_{i,t} = \frac{V_{i,t}}{\mathrm{MA}_N(V_{i,t})}.$$

Here,

$$\mathrm{MA}_N(V_{i,t})$$

represents the moving average of volume over the previous N trading days.

If

$$RV_{i,t} > 1,$$

then the current volume is above the stock's recent average.

This may contain more information than the absolute number of shares traded.

7. Why This Does Not Mean Volume Is Useless

It is important to distinguish between a failed signal and a failed variable.

The experiment tested the following hypothesis:

Can raw trading volume, ranked across stocks, directly predict future returns?

The results do not provide strong evidence in favor of this simple hypothesis.

However, this does not imply that the volume variable itself has no value.

Therefore,

The tested volume signal failed, not necessarily the volume variable.

The volume variable can still be investigated through normalized features and interactions with other factors.

8. How Volume Can Be Used Later

Although raw volume was not successful as a standalone factor, it can still play several useful roles in the later portfolio model.

8.1 1. Volume as a Confirmation Signal

Suppose a momentum model identifies a stock with strong positive momentum.

Instead of relying exclusively on the momentum signal, we can check whether the price movement is supported by above-normal volume.

For example:

$$\text{Momentum Signal} = 1$$

and

$$RV > 1.5$$

could indicate that the price movement is accompanied by unusually high trading activity.

Volume would therefore act as confirmation rather than the primary signal.

8.2 2. Relative Volume

Instead of using raw volume, we can construct:

$$RV_t = \frac{V_t}{MA_N(V_t)}.$$

This measures today's volume relative to the stock's recent average.

This normalization makes the feature more meaningful across stocks with different normal trading volumes.

8.3 3. Volume Percentage Change

Another possible feature is the percentage change in volume:

$$\Delta V_t^{\%} = \frac{V_t - V_{t-1}}{V_{t-1}}.$$

This can identify sudden increases or decreases in trading activity.

8.4 4. Abnormal Volume

Abnormal volume can be measured by comparing current volume with a historical benchmark.

For example,

$$AV_t = \frac{V_t - \mathrm{MA}_N(V_t)}{\mathrm{MA}_N(V_t)}.$$

A positive value indicates volume above its recent average, while a negative value indicates volume below its recent average.

8.5 5. Volume Combined with Momentum

One of the main potential applications is combining volume with the momentum factor.

For example, a combined score could take the form

$$S_t = M_t + \lambda V_t^*$$

where

- M_t is a momentum score,
- V_t^* is a normalized volume score,
- λ controls the contribution of volume.

The objective would not necessarily be for volume to predict returns independently. Instead, volume could improve the reliability of another signal.

8.6 6. Liquidity Filter

Volume can also be used as a liquidity filter.

For example, stocks with extremely low trading activity can be excluded from the portfolio.

This can help reduce problems related to:

- poor liquidity,
- large bid-ask spreads,
- market impact,
- difficulty executing trades.

Therefore, volume can serve as a

$$\boxed{\text{Liquidity Filter}}$$

rather than a return-prediction factor.

9. Limitations of the Current Test

The current experiment has several limitations.

1. **Raw volume was used.**

   Absolute volume does not account for differences in the normal trading activity of individual stocks.

2. **Only a simple grouping methodology was tested.**

   More sophisticated transformations such as relative volume, abnormal volume, and volume momentum were not tested.

3. **The relationship may be conditional.**

   Volume may interact with momentum, volatility, market regime, or price movements.

4. **Group averages can hide underlying variation.**

   An average return for a volume group does not show whether the effect is consistent across individual stocks or time periods.

5. **Statistical significance requires further testing.**

   Differences between group returns should not automatically be interpreted as evidence of a genuine predictive relationship.

6. **Transaction costs were not the primary focus.**

   Even a small apparent signal could become unattractive after accounting for transaction costs, spreads, and market impact.

10. What We Learned

The volume experiment was useful even though the original hypothesis was not strongly supported.

The main lessons were:

1. A commonly used market variable does not automatically become a useful standalone trading factor.

2. A factor should be tested empirically rather than assumed to be predictive.

3. Raw volume contains strong stock-specific characteristics.

4. A lack of monotonicity is an important warning sign for a simple ranking strategy.

5. A factor that fails independently can still be useful as a secondary feature.

6. Normalization can make volume more meaningful across different securities.

The negative result is therefore still useful for the research process because it helps determine which simple approaches should not be used blindly.

11. Future Research

The next stage should investigate transformed versions of volume rather than completely discarding the variable.

Potential features include:

- Relative volume
- Volume moving-average ratios
- Volume percentage change
- Volume acceleration
- Abnormal volume
- Volume-price interaction
- Volume combined with momentum
- Volume combined with volatility
- Liquidity-based filters

A particularly useful feature for future testing is:

$$RV_t = \frac{V_t}{MA_{20}(V_t)}.$$

This measures today's volume relative to the stock's recent 20-day average.

Another possibility is to combine price and volume:

$$S_t = \text{Momentum}_t \times f(RV_t),$$

where $f(RV_t)$ represents a volume-confirmation component.

These approaches should be tested independently before being incorporated into the final portfolio model.

12. Final Conclusion

The volume analysis was conducted to determine whether trading volume could be used as a standalone predictor of future stock returns.

The analysis divided observations into volume groups and compared their subsequent 20-day forward returns.

The resulting group returns did not exhibit a clear monotonic relationship with volume.

Higher raw volume did not consistently correspond to higher future returns, and lower raw volume did not consistently correspond to a specific future return outcome.

Therefore, the simple raw-volume ranking approach was not strong enough to justify using volume as a standalone return-prediction factor.

However, the experiment should not be interpreted as evidence that volume has no value.

The more appropriate conclusion is:

> Raw trading volume did not demonstrate a sufficiently clear standalone predictive relationship with future returns in the tested framework. However, volume remains potentially useful as a supporting feature, liquidity filter, or confirmation signal when combined with stronger price-based factors such as momentum.

Thus, volume will not be discarded from the project. Instead, it will be moved from the category of

Primary Standalone Signals

to the category of

Secondary Features and Filters.

The next research stage can therefore investigate whether normalized or abnormal volume, particularly when combined with momentum or other price signals, provides additional predictive information.

Summary of the Experiment

Table 2: Summary of Volume Research

Aspect	**Finding**
Variable tested	Raw trading volume
Prediction horizon	20 trading days
Testing method	Volume-group sorting and forward-return comparison
Main observation	No clear monotonic relationship between volume group and forward return
Standalone signal	Not sufficiently strong or consistent
Main problem	Raw volume is highly stock-dependent and does not measure abnormal activity
Future use	Supporting feature, volume confirmation, normalized volume, and liquidity filter
Recommended next step	Test relative/abnormal volume and interaction with momentum